

PROJECT REPORT FOR
Foundations of Artificial Intelligence

**OPTIMAL SOLAR PANEL TILT &
AZIMUTH DETERMINATION**

A PROJECT REPORT

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ABSTRACT

Solar photovoltaic energy generation depends significantly on the geographical location and orientation of the panel. This project aims to determine the optimal tilt and azimuth angles of a solar PV panel to maximize its predicted annual energy yield. The system accepts a user-provided address, which is converted into latitude and longitude using a geocoding service. These coordinates are then used with the EU Joint Research Centre PVGIS API to obtain annual energy-yield estimates for different panel orientations, with the peak power fixed at 1 kW and system losses at 14%.

The orientation optimization is formulated as a local-search problem, where tilt and azimuth represent the state and the PVGIS annual energy yield serves as the objective function. Hill Climbing and Simulated Annealing are implemented to search for suitable panel orientations and compare their performance. The system also evaluates different locations, including Chennai, Bengaluru, and Leh, to study the effect of geographical location on optimal panel orientation. The project combines artificial intelligence, solar-energy modelling, geographic information, and API-based computation to provide an automated approach for solar PV orientation analysis.

CHAPTER 1

INTRODUCTION

Solar photovoltaic (PV) systems are an important source of renewable energy, and their energy generation depends on factors such as geographical location and panel orientation. The tilt and azimuth angles of a solar panel determine how it is positioned relative to the sun and therefore influence the amount of energy it can generate. Selecting suitable angles manually for different locations can be time-consuming and may not provide the highest possible annual energy yield.

This project focuses on automatically determining suitable tilt and azimuth angles for a solar PV panel at a given location. The user's address is first converted into latitude and longitude using a geocoding service. These coordinates are then provided to the PVGIS API, which supplies predicted annual energy yield for different panel orientations. The problem is formulated as a local-search optimization problem, where the panel's tilt and azimuth represent the search state and the annual energy yield represents the objective function.

Two Artificial Intelligence techniques, Hill Climbing and Simulated Annealing, are used to search for improved panel orientations. Their results are compared to study their behaviour in finding high-yield orientations. The system also considers different locations, such as Chennai, Bengaluru, and Leh, to examine how geographical location affects the optimal panel orientation.

The project therefore combines solar-energy modelling, geographic information, API-based computation, and AI-based local-search optimization to develop an automated approach for solar panel orientation analysis.

CHAPTER 2

LITERATURE REVIEW

Solar photovoltaic (PV) energy generation is influenced by factors such as geographical location, panel tilt, and azimuth angle. Existing studies have investigated solar-panel orientation and used solar simulation tools and energy databases to estimate PV energy generation for different locations. Various optimization techniques, including Genetic Algorithms, Particle Swarm Optimization, and other search-based methods, have also been explored for improving solar-panel orientation.

Conventional approaches commonly evaluate predefined panel orientations or use fixed assumptions to determine suitable angles. Such approaches may involve testing multiple combinations of tilt and azimuth and may have limited adaptation to specific geographical locations. The project documentation identifies an opportunity to approach this problem using AI-based local-search techniques rather than relying only on direct evaluation of predefined orientations.

This project applies Hill Climbing and Simulated Annealing to solar-panel orientation optimization using real energy-yield data obtained from the PVGIS API. The panel orientation is represented as a search state consisting of tilt and azimuth, while the predicted annual energy yield is used as the objective function. The two algorithms are compared to investigate their performance in finding suitable orientations.

The literature and project review therefore indicate that solar-panel orientation optimization is an established research area. The focus of this work is on the practical integration of AI local-search algorithms with real PVGIS energy-yield data and the study of location-dependent optimization across Chennai, Bengaluru, and Leh.

CHAPTER 3

PROPOSED SYSTEM

The proposed system is designed to automatically determine a suitable tilt and azimuth angle for a solar PV panel based on the user's geographical location. Instead of manually testing different panel orientations, the system formulates the problem as a local-search optimization problem and uses Hill Climbing and Simulated Annealing to search for an orientation that provides a high predicted annual energy yield.

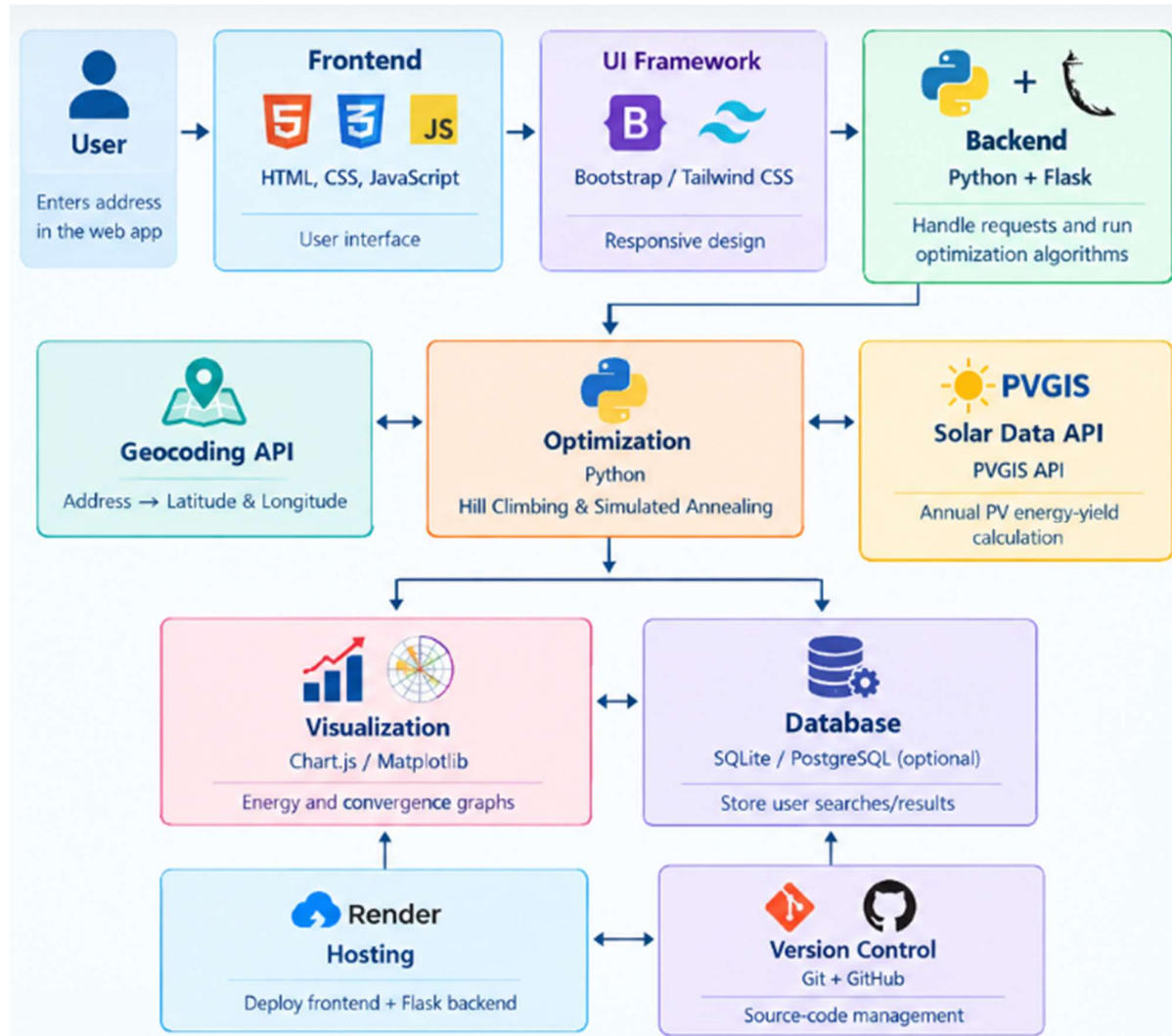
The user enters an address through the web application. A geocoding API converts the address into latitude and longitude, which are then sent to the PVGIS API. PVGIS provides the predicted annual energy yield for the selected panel orientation. The system keeps the peak power at 1 kW and system loss at 14%, while the optimization algorithms determine the tilt and azimuth values.

The optimization process represents (tilt, azimuth) as a state. Hill Climbing evaluates neighbouring orientations and moves towards an improved energy yield. Simulated Annealing explores the search space more widely and can accept lower-yield states during the search, helping it move away from local optima. The results of both algorithms are compared based on the obtained orientation and predicted annual energy yield.

The system also supports comparison across different geographical locations, such as Chennai, Bengaluru, and Leh, to study how location affects the resulting panel orientation. The final output displays the optimized tilt and azimuth along with the corresponding predicted annual energy yield, supported by graphs and charts for visualization.

3.1 ARCHITECTURAL DIAGRAM

The architecture of the proposed system can be represented as follows:



The system accepts a user address through the web interface and uses a geocoding API to obtain its latitude and longitude. The Python-Flask backend sends these coordinates to PVGIS, while the optimization module applies Hill Climbing and Simulated Annealing to determine suitable tilt and azimuth angles. The optimized results and annual energy yield are displayed through graphs and charts. Optional database storage, Render hosting, and GitHub version control support the application.

CHAPTER 4

MODULES DESCRIPTION

The system is divided into four main modules to provide a structured approach for solar-panel orientation optimization. Each module performs a specific function, starting from obtaining the user's geographical location and collecting solar-energy data, followed by optimization of the panel orientation and presentation of the final results. The modules work together to automatically determine a suitable tilt and azimuth angle based on the selected location.

MODULE	DESCRIPTION
Module 1 – Location and Geocoding	Accepts the user's address and obtains the corresponding latitude and longitude.
Module 2 – PVGIS Solar Energy Data	Obtains predicted annual energy yield for different panel orientations using PVGIS.
Module 3 – Orientation Optimization	Uses Hill Climbing and Simulated Annealing to determine suitable tilt and azimuth angles.
Module 4 – Results and Visualization	Displays the optimized orientation, annual energy yield, and graphical results.

4.1 MODULE 1 – LOCATION AND GEOCODING

This module accepts the user's address as input and uses a geocoding API to convert it into geographical coordinates. The obtained latitude and longitude are passed to the subsequent modules for solar-energy calculation.

4.2 MODULE 2 – PVGIS SOLAR ENERGY DATA

This module communicates with the EU Joint Research Centre PVGIS API to obtain

predicted annual solar-energy yield for different panel orientations. The system uses a peak power of 1 kW and system loss of 14%, while tilt and azimuth are provided by the optimization process.

4.3 MODULE 3 – ORIENTATION OPTIMIZATION

This is the main AI module of the system. It represents tilt and azimuth as a state and uses the PVGIS annual energy yield as the objective function. Hill Climbing searches neighbouring orientations for improved yield, while Simulated Annealing explores the search space and provides an opportunity to move away from local optima.

4.4 MODULE 4 – RESULTS AND VISUALIZATION

This module presents the optimization results to the user, including the selected tilt angle, azimuth angle, and predicted annual energy yield. Graphs and charts can be used to display energy yield and algorithm convergence. The results can also be compared across locations such as Chennai, Bengaluru, and Leh.

CHAPTER 5

IMPLEMENTATION AND RESULTS

The system takes a user location, obtains the corresponding geographical coordinates, retrieves solar-energy data from the PVGIS API, and applies Hill Climbing and Simulated Annealing to determine suitable panel tilt and azimuth angles. The results are then presented using suitable visualizations.

5.1 EXPERIMENTAL SETUP

The system is implemented using Python as the primary programming language. A geocoding API is used to convert the user's address into latitude and longitude, while the EU Joint Research Centre PVGIS API provides the predicted annual energy yield for different panel orientations. The Requests library is used for API communication and JSON data processing.

For the experiments, the panel peak power is fixed at 1 kW and system losses are fixed at 14%. The optimization variables are the tilt angle and azimuth angle. Hill Climbing searches neighbouring orientations for an improved energy yield, while Simulated Annealing explores the search space using probabilistic acceptance of candidate states.

The algorithms are evaluated for different geographical locations, particularly Chennai, Bengaluru, and Leh, to study how the optimal panel orientation varies with location. The annual energy yield obtained from PVGIS is used as the primary measure for comparing the orientations and optimization results.

5.2 RESULTS

The system produces the optimal tilt angle, azimuth angle, and predicted annual energy yield for the selected location. The results obtained from Hill Climbing and Simulated Annealing can be compared to examine the behaviour of the two local-search algorithms. The project also compares the optimized orientations for Chennai, Bengaluru, and Leh to investigate location-dependent differences in panel orientation.

The results can be presented using an energy-yield versus tilt-angle graph, algorithm convergence graphs, and a comparison chart for the selected cities. These visualizations help demonstrate the search process and the relationship between geographical location, panel orientation, and predicted annual energy yield. The final system displays the optimized orientation and corresponding energy yield through the application interface.

CHAPTER 6

CONCLUSION AND FUTURE WORK

The project successfully presents an AI-based approach for determining suitable solar-panel tilt and azimuth angles for a given geographical location. By combining geocoding, PVGIS solar-energy data, and local-search optimization, the system reduces the need for manually testing multiple panel orientations. Hill Climbing and Simulated Annealing are used to search for orientations that provide high predicted annual energy yield. The system also enables comparison of optimized orientations across different locations such as Chennai, Bengaluru, and Leh, demonstrating the importance of geographical location in solar-panel orientation.

As future work, the system can be extended with additional optimization techniques and more comprehensive location comparisons. The web application can also be enhanced with improved visualization, database support for storing previous searches, and cloud deployment for easier access. Further experiments can be conducted using additional locations and different optimization parameters to study the behaviour of the algorithms under different conditions.

REFERENCES

1. Benghanem, M. (2011) 'Optimization of tilt angle for solar panel: Case study for Madinah, Saudi Arabia', *Applied Energy*, Vol.88, No.4, pp.1427-1433.
2. Chang, Y.P. (2010) 'Optimal the tilt angles for photovoltaic modules using PSO method with nonlinear time-varying evolution', *Energy*, Vol.35, No.5, pp.1954-1963.
3. Hartner, M., Ortner, A., Hiesl, A. and Haas, R. (2015) 'East to west – The optimal tilt angle and orientation of photovoltaic panels from an electricity system perspective', *Applied Energy*, Vol.160, pp.94-107.
4. Huld, T., Müller, R. and Gambardella, A. (2012) 'A new solar radiation database for estimating PV performance in Europe and Africa', *Solar Energy*, Vol.86, pp.1803-1815.
5. Hussein, H.M.S., Ahmad, G.E. and El-Ghetany, H.H. (2004) 'Performance evaluation of photovoltaic modules at different tilt angles and orientations', *Energy Conversion and Management*, Vol.45, Nos.15-16, pp.2441-2452.
6. Jacobson, M.Z. and Jadhav, V. (2018) 'World estimates of PV optimal tilt angles and ratios of sunlight incident upon tilted and tracked PV panels relative to horizontal panels', *Solar Energy*, Vol.169, pp.55-66.
7. Jamil, B., Siddiqui, A.T. and Akhtar, N. (2016) 'Estimation of solar radiation and optimum tilt angles for south-facing surfaces in Humid Subtropical Climatic Region of India', *Engineering Science and Technology, an International Journal*, Vol.19, pp.1826-1835.
8. Kirkpatrick, S., Gelatt, C.D. and Vecchi, M.P. (1983) 'Optimization by Simulated Annealing', *Science*, Vol.220, No.4598, pp.671-680.
9. Klein, S.A. (1977) 'Calculation of monthly average insolation on tilted surfaces', *Solar Energy*, Vol.19, No.4, pp.325-329.
10. Qadeer, A., Parvez, M., Khan, O., Jafri, H.Z. and Lal, S. (2024) 'Optimization of solar radiation on tilted surface in isotropic and anisotropic atmospheric conditions', *Next Research*, Vol.1, No.2, Article 100075.